

IVIVC App User Guide

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1. Introduction

Establishing an in vitro-in vivo correlation (IVIVC) can support the evaluation of medical devices that include absorbable polymers by relating in vivo changes in material properties over time to corresponding in vitro test results, such as degradation profiles. Potential uses of IVIVCs include evaluating manufacturing or material changes, supporting predictions of in vivo performance when modifying an existing device design, and reducing reliance on animal studies during preclinical development. This tool is intended to facilitate IVIVC development. The user supplies paired in vitro and in vivo fitting data; an optional in vitro prediction dataset can also be supplied. The app applies selected preprocessing steps, fits and compares multiple correlation approaches and model families, evaluates selected performance metrics and cross-validation results, generates fitting and prediction plots, and produces report-ready summaries for user review and external documentation. The app provides information to support user review of models. It does not select, recommend, validate, or establish regulatory acceptability for any model. The app and user guide are advisory; they do not establish requirements, acceptance criteria, or a prescribed model-selection process. Users determine whether and how to document their rationale for selecting a model.

This guide describes the app and offers considerations for reviewing its outputs. It does not establish requirements, acceptance criteria, or a prescribed model-selection process. Users determine how to analyze and interpret their data for the intended use.

For general background on credibility assessment and reporting of computational modeling studies in medical device submissions, see FDA guidance on computational modeling credibility and reporting [1, 2].

For additional discussion of model performance, cross-validation results, uncertainty, residual diagnostics, and model selection considerations, see Appendix A. For an example of interpreting model performance, see Appendix B.

2. What the app does

- Applies selected normalization, scaling, and interpolation options to prepare fitting and prediction data.
- Fits user-selected IVIVC approaches and model families to paired in vitro and in vivo data.
- Calculates selected goodness-of-fit, model-comparison, and cross-validation metrics.
- Generates fitting, comparison, and prediction plots with uncertainty displays where supported.
- Creates summaries for user-selected final approaches or models.

3. Inputs

The IVIVC app requires paired in vitro and in vivo degradation data, along with user-selected analysis settings that define how the data are processed, modeled, and summarized. At a high level, the inputs include:

- a spreadsheet containing in vitro and in vivo time-course degradation data for fitting correlations;
- optionally, a spreadsheet containing in vitro data from which to generate predictions using fitted correlations;
- selection of one or more IVIVC modeling approaches;
- selection of candidate models to evaluate, where applicable;
- optionally, preprocessing options, such as normalization, interpolation, and scaling;
- selected performance metrics to display in the results and reports;
- optionally, cross-validation settings for applicable approaches; and
- grid-search settings for model initialization, where the options can usually be left to their default values.

The sections below provide additional detail on input file structure, required data columns, preprocessing options, model selection, and optional analysis settings.

3.1 Input datasets

The app uses separate fitting and prediction datasets. The fitting dataset is required because it provides the paired in vitro and in vivo data used to establish the IVIVC relationship. The prediction dataset is optional and is used only when applying a fitted relationship to new in vitro observations in order to predict the corresponding in vivo behavior.

In this guide, **response value** means the measured y-axis value that changes with time, such as mass remaining, molecular weight, strength, or another degradation-related endpoint. In the input files, these are the in vitro and in vivo value columns.

Input	Description
Fitting dataset	Required four-column dataset used to establish the in vitro/in vivo relationship. Columns are interpreted as in vitro time, in vitro response value, in vivo time, and in vivo response value.
Prediction dataset	Optional two-column in vitro dataset used after fitting to generate in vivo predictions. Columns are interpreted as in vitro time and in vitro response value.

Supported formats are CSV and Excel (.xls, .xlsx). For Excel workbooks, select the sheet containing the relevant data. For CSV files, the first columns are read directly in the order described above.

3.2 Analysis approaches and models

In this app, an analysis **approach** is the overarching method used to establish or evaluate an in vitro/in vivo correlation. A **model** is the mathematical form used within an approach when an equation is needed to describe the relationship.

3.2.1 Approaches

Approaches differ in how the in vitro and in vivo time courses are related to one another.

Approach	Description	Available models
Approach 1: Direct Correlation	Approach 1 directly relates matched in vitro and in vivo time points after preprocessing (i.e., in vivo time is modeled as a function of in vitro time). The fitted model is then used as a time-scaling function that maps an in vitro time to a predicted in vivo time. This approach can be useful when the two datasets have matching time points, or when interpolation can create matched pairs, and the main question is how in vitro time corresponds to in vivo time.	Double Exponential, Exponential, Linear, Linear (zero intercept), Power
Approach 2: Independent Model Fitting	Approach 2 fits the same model family (e.g., exponential decay) separately to the in vitro and in vivo datasets. With fitting data alone, the report summarizes the two fitted models and uncertainty for those fitted curves. When prediction data are supplied, predictions can be generated by comparing the fitted in vivo and in vitro responses or by comparing fitted characteristic time constants when those values are available. This approach can be useful when the two datasets were measured on different time schedules, when each dataset is expected to follow the same type of degradation curve, or when a specific model form is scientifically meaningful for both datasets.	Double Exponential, Exponential, Mass Loss (Gompertz), Mass Loss (One-Phase), Mass Loss (Surface Erosion), Mass Loss (Two-Phase), Power, Stretched Exponential
Approach 3: Direct Mapping	Approach 3 uses direct interpolation-based mappings between datasets rather than fitting a parametric model. It can describe response-value ratios over time and time ratios across response-value ranges. When prediction data are supplied, those mappings can be used to map in vitro response values or in vitro times to the corresponding in vivo response values or times. This approach can be useful for exploratory comparisons, for datasets where a simple equation does not describe the observed pattern well, or when the user wants a direct data-based mapping rather than a fitted model.	No parametric model selection

3.2.2 Models

Candidate models are the mathematical equations available for fitting within the applicable approaches. Different model forms represent different assumptions about the shape of the degradation relationship, such as linear behavior, power-law behavior, or sigmoidal transition behavior.

Model	Used in	Functional form	Notes
Double Exponential	Approach 1, Approach 2	$y = ae^{bx} + ce^{dx}$	Sum of two exponential terms. It can represent profiles with two apparent rates, but the added flexibility may require more data to estimate stably.

Model	Used in	Functional form	Notes
Exponential	Approach 1, Approach 2	$y = ae^{bx}$	Two-parameter exponential relationship. It can describe monotonic change that accelerates or decelerates exponentially over the fitted range.
Linear	Approach 1	$y = a + bx$	Straight-line empirical relationship with an intercept and slope. It is appropriate when the in vitro/in vivo relationship is approximately linear over the fitted range.
Linear (zero intercept)	Approach 1	$y = ax$	Straight-line empirical relationship constrained to pass through zero. It may be useful when a zero in vitro value should correspond to a zero in vivo value under the selected preprocessing and response scale.
Mass Loss (Gompertz)	Approach 2	$y = y_0 - A \exp\{-\exp[(er_{max}/A)(t_{lag} - x) + 1]\}$	Mass-remaining model based on the modified Gompertz curve. It is intended for asymmetric sigmoidal mass-loss profiles with an apparent lag or plateau, a maximum loss-rate region, and a terminal loss extent.
Mass Loss (One-Phase)	Approach 2	$y = y_0 - A/(1 + e^{-(x-t_{50})/s})$	Mass-remaining model with a delayed sigmoidal loss phase. It is intended for mass-loss profiles that begin constant and show a later drop.
Mass Loss (Surface Erosion)	Approach 2	$y = y_0 \max(1 - kx, 0)^n$	Mass-remaining model for surface-controlled erosion. The fitted exponent n allows the profile shape to adapt to slab-like, cylindrical, spherical, or empirical geometry-like erosion behavior.
Mass Loss (Two-Phase)	Approach 2	$y = y_0 - B(1 - e^{-k_b x}) - D/(1 + e^{-(x-t_{50})/s})$	Mass-remaining model with an optional early burst-loss phase followed by a delayed sigmoidal loss phase. It is intended for mass-loss profiles that may show an initial drop, an intermediate plateau, and a later drop.

Model	Used in	Functional form	Notes
Power	Approach 1, Approach 2	$y = ax^b$	Power-law empirical relationship. It can describe nonlinear monotonic relationships where the response changes as a fitted power of time or the mapped predictor.
Stretched Exponential	Approach 2	$y = ae^{bx^c}$	Exponential relationship with a fitted stretching exponent. It can describe non-single-rate behavior, but the additional shape parameter can increase sensitivity when data are sparse.

Mass loss model notes

The surface erosion mass loss form is based on geometry-controlled surface erosion concepts from the erodible slab, cylinder, and sphere model [3]. The one-phase mass loss model uses a logistic/sigmoidal transition to represent a delayed loss phase [4]. The two-phase mass loss model combines an early burst-like loss term with a delayed sigmoidal loss phase. Burst and delayed/multiphase behavior in degradable polymer systems is discussed in the polymer erosion and PLGA release literature [5, 6, 7]. The Gompertz mass loss model uses a modified Gompertz-style sigmoidal form with lag and maximum-rate parameters [8].

3.3 Preprocessing options

Preprocessing prepares the uploaded data for comparison, fitting, and prediction. Normalization expresses response values relative to a reference value so datasets can be compared on a common relative scale. Scaling changes the numerical scale of time or response values without changing the underlying pattern. Interpolation estimates response values at time points that were not directly observed and may be useful when an approach requires matching in vitro and in vivo time points (i.e., Approach 1, Approach 3).

3.4 Normalization

Option	Internal key	Description
Min-max normalization	min_max	Converts response values to a 0-1 scale using the observed minimum and maximum within each dataset.
Initial-value normalization	initial_value	Expresses each response value relative to the first response value in the corresponding dataset.

3.5 Scaling

Option	Internal key	Description
Log10 time scaling	log_x	Uses a base-10 log scale for time values before fitting or interpolation.
Log10 response scaling	log_y	Uses a base-10 log scale for response values before fitting or interpolation.

3.6 Interpolation

Option	Internal key	Description
No interpolation	none	Uses only the original measured points. Response values between measurements are not estimated, so unmatched positions are marked as missing.
Shared-time interpolation	default	Uses linear interpolation to estimate response values for both datasets at a shared set of experimental time points. Estimates outside the measured time range are marked as unavailable.
Densified interpolation	alternative	Adds extra evenly spaced points between adjacent observations before interpolation. This can smooth displays and make fitting more stable, but the added response values are estimated rather than directly measured and can affect metric values.

3.7 Performance metrics

Performance metrics are summary measures that help describe how closely a fitted model matches the observed data (e.g., R^2), how large the prediction errors are (e.g., RMSE), and, for some metrics, how well a model balances fit quality with model complexity (e.g., information criteria). The following performance metrics are included in the app.

Metric	Internal key	Preferred direction	Description
R^2	r_squared	Higher is better	Coefficient of determination. Higher values indicate that the fitted curve follows more of the observed response pattern.
Adjusted R^2	adjusted_r_squared	Higher is better	R^2 adjusted for the number of fitted parameters. It can decrease when extra parameters do not improve the fit enough to justify the added complexity.
MSE	mse	Lower is better	Average squared difference between observed and fitted values. Lower values indicate smaller model errors, but the squared units can be harder to interpret directly.
RMSE	rmse	Lower is better	Typical model error in the same response units as the fitted data. Lower values indicate predictions closer to the observations.
MAE	mae	Lower is better	Average absolute model error in the same response units as the fitted data. Lower values indicate predictions closer to the observations.
NRMSE (mean)	mnrms	Lower is better	RMSE divided by the response mean. Lower values indicate smaller error relative to the typical response magnitude.
NRMSE (std. dev.)	nrmse	Lower is better	RMSE divided by the response standard deviation. Lower values indicate smaller error relative to the amount of variation in the response.

Metric	Internal key	Preferred direction	Description
AIC	aic	Lower is better	Akaike Information Criterion. AIC compares fitted models by considering both fit quality and the number of fitted parameters. Among comparable fitted models, a lower AIC value generally indicates better empirical support for a model that fits the data without unnecessary complexity. Absolute AIC values, including negative values, are not standalone quality grades.
AIC _c	aicc	Lower is better	Corrected Akaike Information Criterion. AIC _c is AIC with an added small-sample correction that becomes increasingly relevant when the number of fitted observations is limited relative to the number of fitted parameters. See the user guide for additional information. Among comparable fitted models, a lower AIC _c value generally indicates better empirical support for a model that balances fit quality and complexity; AIC _c is undefined when there are too few observations relative to the number of parameters.
BIC	bic	Lower is better	Bayesian Information Criterion. BIC compares fitted models by considering both fit quality and the number of fitted parameters, usually applying a stronger penalty for extra parameters than AIC. Among comparable fitted models, a lower BIC value generally indicates better empirical support for a model that fits the data without unnecessary complexity. Absolute BIC values, including negative values, are not standalone quality grades.

For additional background on common error metrics, scale-dependent accuracy measures, and adjusted R^2 , see references [9, 10, 11].

AIC_c adds a small-sample correction to AIC that is relevant when the ratio of fitted observations to estimated parameters is small. A commonly cited rule of thumb is $n/k < 40$, where n is the number of fitted observations and k is the number of model parameters [12]; this condition is common in typical IVIVC datasets.

By default, the app selects R^2 , RMSE, and AIC_c. This default set has been useful in internal testing with real datasets, but it is not intended as a universal metric set. Additional metrics are available when they provide useful context for a specific analysis. The default set gives three complementary views of model performance. R^2 is widely known and describes how well the fitted curve follows the observed response pattern. RMSE reports typical error in the original response units, making it easier to judge whether the error is practically large or small. AIC_c supports comparison among fitted candidate models by balancing fit quality with a penalty for unnecessary model complexity, with a correction that is useful for smaller datasets.

3.8 Cross-validation options

Cross-validation provides an estimate of the model's ability to predict observations not used for fitting. The app does this by repeatedly setting aside part of the dataset, fitting the model to the remaining observations, and comparing

model predictions with the observations that were set aside. When the cross-validation error is similar to the error from fitting the full dataset, this suggests the fitted relationship is reasonably stable.

In this app, CV is intended as a **prediction-error diagnostic** and sensitivity check, not as a confidence interval or parameter uncertainty estimate. If the error from the model fitted to the full dataset is similar to the CV error, that provides reassurance that the fitted relationship is not overly dependent on a particular subset of observations. If the CV error is much worse, the fitted relationship might be sensitive to the available data, the selected splitting scheme, or the number and placement of time points.

For time-course data, the splitting scheme matters because nearby observations can contain related information, and early, middle, and late time points can represent different parts of the degradation process. Setting aside a random subset is a general check of whether the fitted curve captures the overall response pattern, but it is less time-aware. Setting aside one time point mainly asks whether the model can recover an individual omitted observation. Setting aside a contiguous block asks whether the model can bridge a missing region of the profile.

Shuffle split is the default CV scheme because it provides a general robustness check of whether the selected curve form captures the overall degradation or correlation pattern. It is not a time-aware forecasting test. Use leave-contiguous-block-out when sensitivity to a missing time region is the main concern, and use leave-one-timepoint-out when the dataset is too small to set aside larger groups of observations. When results change substantially across CV schemes, review the fitted plots, residual diagnostics, parameter uncertainty, model complexity, and scientific plausibility rather than treating CV as a pass/fail result.

The default CV settings have been useful in internal testing with real datasets but are not intended to be universally applicable. Possible reasons to adjust them include small datasets or unstable cross-validation results.

For background on cross-validation procedures and their use in model assessment and selection, including issues that arise with time series, see references [13, 14, 15, 16].

3.8.1 Choosing a cross-validation scheme

The table below summarizes the schemes available in the app. The “CV error describes” column explains what kind of prediction sensitivity each scheme is most directly measuring.

Scheme	CV error describes	Useful when	Main caution
Shuffle split	Prediction error for repeated random subsets of observations set aside from fitting.	Default general robustness check for whether the fitted curve captures the overall response pattern.	Not time-aware; can appear overly reassuring when nearby time points carry similar information.
Leave-one-timepoint-out	Prediction error when one time point is omitted at a time; sensitivity to setting aside one observation.	Useful for very small datasets because each repeated fit still uses nearly all of the available data.	Can be a gentle test when neighboring time points have similar response values to the omitted point.
Leave-contiguous-block-out	Prediction error when a block of adjacent time points is omitted; sensitivity to a gap in the observed profile.	Primary time-aware sensitivity check, especially when a missing time region is a practical concern.	Larger blocks make the test harder but can leave too few points for the model to fit stably. Start with small blocks and increase cautiously.

Scheme	CV error describes	Useful when	Main caution
None	No cross-validation is performed; only the model fitted to the full dataset is summarized.	Useful when there are too few observations to split meaningfully, repeated fits are numerically unstable, or the goal is simply to examine the fitted model itself.	Removes one diagnostic tool. When CV is skipped, place greater emphasis on fitted plots, residuals, final-fit metrics, parameter diagnostics, and scientific plausibility.

3.8.2 Cross-validation parameters

Parameter	Internal key	Description
Cross-validation scheme	cv_scheme	Selects the splitting method used to estimate cross-validation performance for fitted parametric models. Shuffle split is the default general robustness check.
CV splits/block size	cv_n_splits	For Shuffle split, the number of repeated random train/test splits. For Leave-contiguous-block-out, the number of adjacent time points set aside per split. The setup page does not enforce a maximum block size because the uploaded data are not read until Analyze is selected; larger blocks may leave too few observations for stable fitting. This setting is not used for Leave-one-timepoint-out or None.
Shuffle split test size	cv_test_size	The fraction of observations used as test data in each Shuffle split repeat. Smaller values leave more observations for fitting; larger values make the test more demanding but leave fewer observations for training.
CV random seed	cv_random_state	Seed used to make randomized cross-validation splits reproducible. This applies to Shuffle split.

3.9 Grid-search initialization

When the app fits a mathematical curve, it needs starting values for the model parameters. Poor starting values can lead to an unstable or unreasonable fit, especially for nonlinear models. Grid-search initialization reduces this risk by trying many candidate starting-value sets before nonlinear least-squares fitting.

The app evaluates those candidate starts, chooses the most promising one, and then runs the final curve-fitting step from that starting point. This improves robustness, but it does not establish that the fit is the global optimum.

The default grid-search settings can usually be left unchanged. If fitting fails or appears unstable, consider increasing the number of random starts, adjusting the candidate starting-value range, changing the random seed, or reviewing whether the selected model form is suitable for the data.

Parameter	Internal key	Description
Grid-search random starts	grid_search_num_points	Number of candidate starting parameter sets evaluated before nonlinear least-squares fitting. Higher values may improve robustness but increase runtime.
Grid-search CPU cores	grid_search_num_cores	Number of CPU cores used for the starting-point search. Only available via command-line interface.

Parameter	Internal key	Description
Grid-search parameter minimum	grid_search_param_min	Lower bound for randomly sampled candidate starting parameter values.
Grid-search parameter maximum	grid_search_param_max	Upper bound for randomly sampled candidate starting parameter values.
Grid-search random seed	grid_search_random_state	Seed used to make randomized starting-point sampling reproducible. Changing the seed changes the sampled candidates while keeping the same bounds and number of starts.

3.10 Residual diagnostic display

Raw residual (i.e., prediction error) plots and normal quantile-quantile (Q-Q) plots are selected by default in the Advanced Analysis section. Either plot type can be hidden before analysis when a more compact results view is preferred. Raw residual plots show observed-minus-predicted values in the original response units. Each plot uses its own symmetric y-axis centered at zero, with a minimum display range relative to the observed response range so very small residuals are not visually exaggerated. Q-Q plots are omitted when residual variation is too small relative to the response scale to make the diagnostic informative.

4. Outputs

The IVIVC app generates interactive results and downloadable reports that summarize fitted model behavior, model-comparison metrics, diagnostic information, and user-selected analysis settings. Output availability depends on the selected approaches, whether fitting succeeds, and whether a prediction dataset was supplied.

Output	Description
Preprocessing outputs	Processed data tables and plots showing the data used for fitting and prediction.
Model fitting outputs	Parameter estimates reported as estimate $\pm$ 95% confidence interval where available, fitted plots with error bars, residual plots, and selected performance metrics.
Model comparison outputs	Tables comparing selected approaches, models, metrics, relative model evidence, and cross-validation summaries.
Prediction outputs	Prediction plots and downloadable tables when a prediction dataset is supplied and the selected method can be applied.
Selected model & summary of outputs	User-selected report summaries and optional report export for incorporation into external documentation.

4.1 Relative model evidence

Relative model evidence tables compare candidate parametric models using information criteria (i.e., AIC, AIC_c, and/or BIC), showing which models have more support from the observed data after balancing fit quality against model complexity. These comparisons are only meaningful among models fitted to the same data, with the same preprocessing settings, and within the same IVIVC approach.

These tables are shown only when one or more information criteria are selected as metrics. Interpret each selected criterion separately—AIC, AIC_c, and BIC can occasionally rank the same models differently. See the appendix for interpretation guidance, rules of thumb, and limitations.

The relative model evidence tables contain the following quantities.

Table output	What it means
n	Number of observations used to fit the model.

Table output	What it means
k	Number of fitted model parameters. Models with more parameters can fit more flexible shapes, so information criteria include a complexity penalty.
AIC, AIC _c , or BIC	Information-criterion value for the fitted model. Lower values indicate stronger relative support within the candidate set.
Delta	Difference between a model's criterion value and the lowest value in the candidate set. Values near 0 indicate similar support to the leading model; larger values indicate less support.
Weight	Normalized relative-support score across the candidate set. Weights are comparative summaries, not absolute probabilities that a model is correct.
Evidence ratio	Ratio of the highest weight in the candidate set to the model's weight. Larger values indicate less relative support compared with the leading model.
Support badge	Rule-of-thumb label based on the delta value. Intended to aid user review, not to recommend a model.

4.2 Uncertainty and error bars

These displays show how much fitted curves or predictions vary based on the fitted model and residual error. They are intended to support review of uncertainty, not to define acceptance limits.

Very large uncertainty bands can extend beyond the initial plot range and remain out of view until the plot is rescaled. When this occurs, the app displays an informational badge near the affected fitting plot. A separate warning is shown if one or more uncertainty bounds are nonfinite or the uncertainty band cannot be calculated. The badge describes the display condition but does not identify its cause; review the fitted parameter diagnostics, model form, and available data when it appears.

The following descriptions summarize how fitted-parameter uncertainty, fitting-plot bands, and prediction error bars are calculated or interpreted for each approach.

4.2.1 Approach 1: Direct Correlation

Fitting uncertainty

Parameter estimates: fitted parameters are reported as estimate $\pm$ 95% confidence interval. The plus-minus value is calculated from the fitted model's covariance matrix, parameter standard errors, and residual degrees of freedom. It summarizes how precisely each parameter was estimated from the available data.

Fitting plot bands: shaded 95% prediction bands are generated by Monte Carlo sampling of the fitted parameter covariance matrix. The sampled model variance is combined with the residual variance from the fitted data, then converted to a two-sided interval using the appropriate Student's t multiplier when residual degrees of freedom are available.

These bands describe approximate prediction uncertainty around the fitted relationship, not regulatory acceptance limits. They depend on the fitted model form, local covariance approximation, residual behavior, and preprocessing choices.

Prediction uncertainty

Prediction plot error bars: when a prediction dataset is supplied, predicted in vivo times are plotted with horizontal error bars. The bars are calculated from the same 95% prediction-band calculation used for the fitted correlation model; the upper and lower bar lengths are the distances from the band mean to the corresponding upper and lower interval bounds at each input prediction time.

The error bars therefore include propagated fitted-parameter uncertainty and residual/model error from the Approach 1 correlation fit.

4.2.2 Approach 2: Independent Model Fitting

Fitting uncertainty

Parameter estimates: each in vitro and in vivo model parameter is reported as estimate $\pm$ 95% confidence interval. The plus-minus value is calculated from the corresponding fitted model's covariance matrix, parameter standard errors, and residual degrees of freedom. It summarizes how precisely each parameter was estimated from the available data.

Time constants: tau values, when available, are calculated from the fitted model and parameter covariance. Reported tau values are shown as tau $\pm$ 95% confidence interval. The interval accounts for uncertainty in the fitted parameters used to calculate the time constant.

Fitting plot bands: the in vitro and in vivo shaded 95% prediction bands are generated separately. For each fitted model, the app Monte Carlo samples the fitted parameter covariance matrix, combines sampled model variance with residual variance, and converts the resulting standard uncertainty to a two-sided interval using the appropriate Student's t multiplier when residual degrees of freedom are available.

Prediction uncertainty

Response-value-ratio prediction error bars: uncertainty is propagated from both fitted models. The app samples each model's parameter covariance matrix, calculates the in vivo/in vitro response-value ratio for each sample, combines variance in that ratio with residual-based ratio uncertainty, and converts that uncertainty to a two-sided 95% interval. The plotted vertical error bars are scaled from the resulting ratio interval at each prediction time.

Time-constant-ratio prediction error bars: when both tau values are valid, uncertainty from both tau values is carried through the ratio calculation. The resulting uncertainty in the scaled prediction time is converted to a symmetric 95% interval, where possible, and plotted as a horizontal error bar.

Response-value ratio uncertainty can become unstable when the denominator model prediction or denominator time constant is near zero. Predictions in those regions may be highly sensitive to small changes in the fitted models.

4.2.3 Approach 3: Direct Mapping

Fitting uncertainty

Parameter uncertainty: Approach 3 does not fit a parametric model, so there is no fitted-parameter covariance matrix, no parameter confidence intervals, and no model-based 95% prediction band.

Fitting plots: the plots show interpolation-based ratios/mappings calculated from the preprocessed datasets. They represent deterministic mappings from the supplied data and preprocessing choices, not statistical uncertainty bands.

Prediction uncertainty

Prediction plot error bars: Approach 3 does not generate formal propagated error bars for prediction outputs because no parametric covariance model is fitted.

Uncertainty is therefore limited to the quality of the input data, preprocessing, interpolation, and the assumption that the observed mapping remains valid for the prediction dataset. Extrapolated points outside the interpolation range are excluded or returned as unavailable.

5. Miscellaneous

5.1 Command-line interface

The repository includes `cli.py` for batch-oriented fitting from the command line. Use `python cli.py --help` to see available arguments for input files, approaches, models, metrics, preprocessing options, cross-validation settings, and grid-search settings. The internal keys specified in this user guide are used to specify settings with `cli.py`.

5.2 Extending the app

The IVIVC app is designed so that common extensions can be added with minimal changes to the main application code. In many cases, new models, performance metrics, and preprocessing methods can be added by following the structure of the existing files in the relevant subdirectories.

5.3 Data handling

The app does not save uploaded datasets or fitting results to persistent storage. Uploaded data are processed for the current analysis, and generated results and downloads are returned directly to the user.

Appendices

A. How to Interpret Model Performance

No single metric determines whether an IVIVC model is meaningful. An evaluation of model performance can include goodness of fit, prediction error, model complexity, cross-validation behavior, residual patterns, parameter uncertainty, and whether the fitted relationship is scientifically plausible for the polymer degradation mechanism and intended use. Performance metrics are descriptive and comparative tools, not acceptance criteria. Small datasets can produce unstable estimates, and similar metric values do not necessarily indicate equivalent scientific validity. For analyses intended to support regulatory submissions, the depth of supporting evidence depends on the model's intended use and applicable credibility considerations [1, 2].

The app reports metrics and comparison tables to support user review. It does not select, recommend, validate, or establish regulatory acceptability for any model. Users determine whether and how to document preprocessing assumptions, fitting settings, model selection rationale, limitations, and any planned use of the model outside the fitted data range.

A.1 Model comparability before interpreting metrics

Performance metrics are most meaningful when the models being compared were evaluated under identical conditions. Metric comparisons are most interpretable within the same uploaded dataset, preprocessing configuration, response scale, selected IVIVC approach, and model set.

In particular, AIC, AIC_c, and BIC values are not comparable across different datasets, transformed responses, preprocessing settings, or substantially different fitting contexts. These information criteria measure the *relative* support among models evaluated under the same conditions—they are not absolute quality scores and lose their meaning when the underlying data or setup differs. Approach 3 is interpreted differently from Approaches 1 and 2 because it uses direct interpolation-based mappings rather than fitting a parametric model.

A.2 Metric ranges, direction, and interpretation

Metric	Range / direction	Interpretation guidance
R^2	Best possible value is 1. Values near 0 indicate little improvement over predicting the mean response. Can be negative if the model performs worse than the mean-response baseline. Higher is better.	Proportion of response variance explained by the model [17]. High R^2 does not establish that the model is mechanistic, causal, predictive outside the fitted data range, or suitable for the intended use.
Adjusted R^2	Higher is better. Can be lower than R^2 .	Penalizes additional fitted parameters, making it more informative than R^2 when comparing models of different complexity [11].
MAE	Lower is better. Units match the response units.	Mean absolute error on the response scale [17]. Interpret relative to assay variability, measurement uncertainty, and the intended use.
MSE	Lower is better. Units are squared response units.	Mean squared prediction error [9, 17]. Useful computationally, but less intuitive than RMSE because the units are squared.
RMSE	Lower is better. Units match the response units.	Root mean squared error, the typical prediction error on the response scale [9, 17]. Interpret relative to assay variability, measurement uncertainty, and the intended use. Penalizes larger errors more heavily than MAE.
NRMSE	Lower is better. Normalized to allow more meaningful comparison across response scales.	RMSE prediction error expressed relative to a normalization factor such as the response mean or standard deviation. Interpretation depends on the normalization method used.

Metric	Range / direction	Interpretation guidance
AIC	Lower is better, but only relative to other comparable models. No fixed bounds; negative values are allowed.	Balances goodness of fit against model complexity [18, 12, 19]. A lower value indicates a model that explains the data well without unnecessary parameters. Absolute values are not standalone quality grades.
AIC _c	Lower is better, but only relative to other comparable models. Undefined when there are too few observations relative to the number of fitted parameters.	A small-sample correction to AIC that is most relevant when the observation-to-parameter ratio is low [20, 12, 19]; $n/k < 40$ is a rule of thumb [12].
BIC	Lower is better, but only relative to other comparable models. No fixed bounds; negative values are allowed.	Balances goodness of fit against model complexity, typically applying a stronger penalty for additional parameters than AIC [21, 12, 19].
Cross-validation metrics	Direction follows the underlying metric: higher is better for R^2 -type metrics, lower is better for error metrics.	Estimate prediction performance under the selected splitting scheme [13, 14, 15, 16]. Can help identify overfitting, instability, or sensitivity to individual observations or data splits.
Fit/CV ratio	Values near 1 indicate similar fitted and cross-validation performance.	A ratio near 1 suggests the model generalizes consistently. Substantial deviations can indicate overfitting, instability, or sensitivity to the splitting scheme, though no universal acceptance threshold exists.

A.3 Interpreting goodness of fit and prediction error

Goodness-of-fit metrics— R^2 , adjusted R^2 , MSE, MAE, RMSE, and NRMSE—describe how well the model fits the data used to estimate its parameters. Their interpretation depends on practical context, not just numerical value, and there is no universal IVIVC performance cutoff.

Strong goodness-of-fit is encouraging but can be misleading, particularly for small datasets or models with many parameters. Excellent fit metrics do not rule out unstable parameters, poor cross-validation performance, systematic residual patterns, or a fitted curve shape that is not scientifically plausible for the degradation process being modeled.

RMSE is often the most intuitive error metric because it is expressed in the same units as the response. NRMSE can be useful for comparing error across different response scales, but the choice of denominator changes what the metric actually measures [10]:

- **NRMSE normalized by the response mean** answers: “How large is the typical prediction error relative to the average response value?” This normalization is useful for comparing model error across endpoints with different scales or units. For example, comparing how well a model fits mass loss data versus molecular weight data on a common scale.
- **NRMSE normalized by the response standard deviation** answers: “How large is the typical prediction error relative to the natural spread of the observed responses?” This normalization is useful for gauging how much of the variability in the data the model fails to capture. A value approaching or exceeding 1 indicates that the typical prediction error is as large as the spread of the data itself, suggesting the model explains little of the observed variability, analogous to an R^2 near zero.

Take care when comparing NRMSE values across datasets or response scales if the normalization method differs. More broadly, the practical importance of a given error level depends on context: a numerically small error may still matter when the relevant degradation endpoint has narrow variation, while a larger error may be less consequential during exploratory model development when high predictive precision is not central to the analysis.

A.4 Interpreting relative model evidence

Relative model evidence tables compare candidate models using information criteria (i.e., AIC, AIC_c, and BIC), which balance goodness of fit against model complexity. In this context, *empirical support* means that the observed data provide more support for one model than another within the displayed candidate set, based on the selected information criterion. It does not prove that a model is correct, mechanistic, predictive outside the observed range, or suitable for a regulatory submission.

Rules of thumb for delta values

Delta is the difference between a model's criterion value and the lowest value in the candidate set. Common interpretive conventions are:

- **AIC / AIC_c**: delta 0–2 suggests similar support; 4–7 suggests less support; greater than 10 suggests essentially no support relative to the leading model [12].
- **BIC**: delta 0–2 suggests similar support; 2–6 suggests less support; 6–10 suggests substantially less support; greater than 10 suggests essentially no support relative to the leading model [22].

These are interpretive conventions, not regulatory acceptance criteria.

Evidence weights

For a given criterion, the evidence weight for model i is:

$$w_i = \frac{\exp(-\frac{1}{2}\Delta_i)}{\sum_{j=1}^J \exp(-\frac{1}{2}\Delta_j)} \quad (1)$$

where Δ_i is the delta value for model i and J is the number of candidate models. Larger weights indicate stronger relative support within the candidate set [12, 19].

When multiple models have similar support

If several models have similar delta values or weights, relative support alone is not sufficient to choose among them. Other considerations include model simplicity, parameter plausibility, residual behavior, cross-validation performance, and scientific interpretability.

A.5 Interpreting cross-validation performance

Cross-validation evaluates prediction performance using observations left out when the model was fit. In this app, it is most useful as a **sensitivity diagnostic**: when the cross-validation error is similar to the error from fitting the full dataset, this suggests the fitted relationship is reasonably stable regardless of which observations are set aside. When the two differ substantially, the model may be more sensitive to which specific data points were included in fitting, or to the particular way the data were split. Cross-validation does not measure uncertainty in the model parameters. The practical question to keep in mind is not just “how good is the score?” but also “what kind of prediction task did the cross-validation scheme actually create?”

What each scheme is testing

Different cross-validation schemes answer different questions, and those questions determine how the results are interpreted. Shuffle split, the default, sets aside randomly selected time points without regard to their order. It serves as a general check of overall curve-shape robustness, but it may make the model appear more reliable than it is when nearby time points carry similar information. Leave-one-timepoint-out tests how well the model predicts a single omitted time point when all others are available, a relatively gentle task when neighboring time points are close together. Leave-contiguous-block-out tests whether the model can bridge an entire missing time window, a harder and more time-aware task that is more sensitive to gaps in the profile.

These distinctions matter because adjacent time points in a degradation profile are often similar to one another, measurement errors may be related across time, and early, middle, and late portions of a degradation profile may reflect

qualitatively different behavior. Standard cross-validation procedures are reviewed in [13]; additional considerations for temporally ordered or dependent data are discussed in [14, 15, 16].

Interpreting results for small IVIVC datasets

For the small datasets typical in IVIVC development, cross-validation results are intended as an exploratory diagnostic or sensitivity analysis rather than a definitive performance benchmark. Compare the CV metric with the corresponding metric for the model fitted to the full dataset, but also consider whether the chosen splitting pattern reflects the practical prediction question of interest.

For example, strong leave-one-timepoint-out performance alongside weaker leave-contiguous-block-out performance may indicate that the model can interpolate individual missing points well, but becomes less reliable when an entire local region of the profile is absent from fitting. The importance of that distinction depends on the intended use of the model.

Fit/CV ratios

The app reports fit/CV ratios to summarize the relationship between full-fit and cross-validation performance for the same metric. A ratio near 1 indicates that the two are similar. A ratio far from 1 may suggest reduced generalizability, overfitting, model instability, or sensitivity to the splitting scheme. Because the direction of concern depends on the metric (for example, a ratio above 1 is concerning for RMSE but not for R^2), interpretation of the ratio depends on the raw fit and CV values, the chosen splitting scheme, and the fitted plots.

Cross-validation comparison badges

The app displays cross-validation comparison badges as interpretive aids. Badges are assigned based on the preferred direction of the metric: for lower-is-better metrics such as RMSE, a worse CV result means the CV value is larger than the full-fit value; for higher-is-better metrics such as R^2 , a worse CV result means the CV value is smaller. The thresholds used are:

- **Similar:** CV performance is within a factor of 1.25 of the full-fit value.
- **Moderately worse:** CV performance is 1.25- to 2-fold worse than the full-fit value.
- **Substantially worse:** CV performance is more than 2-fold worse than the full-fit value.

These thresholds are heuristic review aids to help flag results that warrant closer attention. They are not statistical acceptance criteria or regulatory standards.

Metrics excluded from cross-validation comparisons

The app does not include adjusted R^2 , information criteria (AIC, AIC_c, and BIC), or normalized RMSE (NRMSE) in cross-validation comparison tables. The reason is that these metrics are not directly comparable across fits that use different numbers of observations:

- **Adjusted R^2 and information criteria** incorporate the number of observations or fitted parameters as part of their calculation. Because the final model and each cross-validation model are fitted to different subsets of the data, these quantities are not on the same scale and are not directly comparable.
- **NRMSE** is normalized by a factor that generally differs across subsets. This means the final-fit and CV values can have different denominators, making a direct comparison misleading.

Use the eligible cross-validation metrics to assess prediction behavior during CV. The ineligible CV metrics remain available for evaluating and comparing fitted models. Note that not all eligible metrics are available for every cross-validation scheme. Some metrics require more than one observation to be defined or stable. For example, R^2 cannot be calculated for a single omitted point and will be unavailable under leave-one-timepoint-out CV. Where a metric is unavailable for the selected scheme, rely on the metrics that are reported.

A.6 Interpreting residual diagnostics

Residuals are the differences between observed and predicted values (i.e., the errors in the predictions). A well-specified model generally produces residuals without systematic structure. Residual diagnostics help assess whether model errors show systematic patterns [23]. The app provides two qualitative diagnostic plots: residual-vs-time plots and quantile-quantile (Q-Q) plots. If your dataset contains few time points, which is typical of IVIVC datasets, these plots are intended as exploratory diagnostics rather than pass/fail tests. With only a few points, random variation can sometimes appear to form a meaningful pattern.

Residual-vs-time plots

Residuals are plotted in the same units as the fitted response. The axis covers the largest absolute residual for each model but is not allowed to shrink below a small fraction of the observed response range. This minimum prevents extremely small numerical residuals from being expanded to fill the plot and appearing more important than they are. To compare model performance, use an overall error metric like RMSE or MAE rather than residual plots.

Q-Q plots

Quantile-quantile (Q-Q) plots compare the ordered standardized residuals against the pattern expected for randomly distributed errors (i.e., a normal distribution, Gaussian distribution, or ‘bell curve’). Points near the diagonal reference line are broadly consistent with random residual variation. Systematic curvature, large departures in the tails, or isolated extreme points may suggest non-randomness, outliers, or a model form that does not fit the data well. When residual variation is too small relative to the response scale, the app reports that the Q-Q plot is not informative rather than magnifying numerical differences.

What to look for

Residual diagnostics are intended for qualitative review—by eye—rather than use as formal statistical tests. Quantitative tests are available for problems such as non-random errors, correlated errors across time (autocorrelation), and variance that changes across the response range (heteroscedasticity) [23]. However, with the small numbers of observations typical of IVIVC datasets, these tests may have limited ability to distinguish systematic problems from random variation. Visual patterns are prompts for further review rather than definitive findings. Residuals scattered around zero in the residual-vs-time plot, and points near the diagonal in the Q-Q plot, are consistent with a model form that captures the observed pattern, though they do not establish model validity. The following patterns may indicate bias or a model form that does not fully capture the data:

- Sustained curvature or systematic trends in the residuals over time.
- Clustering, where residuals at adjacent time points are consistently on the same side of zero.

When any of these patterns appear, consider whether a different model form or a transformed (i.e., normalized or scaled) response might better describe the underlying relationship.

A.7 Interpreting parameter uncertainty

Model parameters cannot always be estimated precisely from the available data. The app reports each parameter estimate $\pm$ 95% confidence interval. The plus-minus value represents the confidence-interval half-width around the estimate. In practical terms, a wide interval means the data do not pin down that parameter precisely, while a narrow interval means the estimate is more stable.

The interval is calculated from the fitted model’s covariance matrix and residual degrees of freedom. This calculation estimates uncertainty from the behavior of the model near the fitted parameter values. For nonlinear models, especially with small datasets, the true uncertainty can be wider, asymmetric, or more strongly affected by relationships among parameters than the reported interval suggests. Interpret the interval together with the relative standard errors, parameter-coupling diagnostics, fitted curves, and scientific plausibility rather than treating it as a precise boundary for the parameter. For a practical overview of nonlinear regression and interpretation of fitted results, see [24]. For a technical treatment of covariance-based parameter uncertainty and approximate confidence intervals in nonlinear least-squares models, see [25].

The app summarizes parameter uncertainty using informational badges as screening aids, not acceptance criteria. They highlight fitted-parameter uncertainty that may affect interpretation or prediction.

Diagnostic	What it summarizes	Interpretation guidance
Relative standard error	The uncertainty in a parameter estimate expressed as a percentage of the estimate itself.	A large value means the uncertainty is large relative to the estimate, so the data are not pinning down that parameter well. The app flags values of at least 50% as elevated and at least 100% as large.
Parameter coupling / correlation	How strongly pairs of parameter estimates move together during fitting (pairwise correlation).	When two parameters are highly correlated, the model can reach a similar fit by trading off one against the other, making both harder to estimate reliably. Some coupling is normal in nonlinear models. The app reports correlation alone as <i>parameter coupling noted</i> ; when high correlation coincides with elevated relative standard error or a numerical sensitivity flag, it reports a stronger identifiability caution. Absolute correlations of at least 0.90 are noted, with stronger concern when correlations of at least 0.95 occur alongside other uncertainty diagnostics.

If no parameter uncertainty concerns are flagged, the app did not detect large relative standard errors or strong parameter coupling under these screening checks. This does not validate the model, confirm mechanistic correctness, or establish regulatory acceptability. Conversely, a coupling or warning badge does not automatically invalidate a model; rather, it is a prompt to review the flagged parameters alongside the fitted curves, residuals, cross-validation behavior, and scientific plausibility.

Parameter uncertainty deserves particular attention when a model will be interpreted mechanistically, used to extrapolate beyond the observed data range, or used to compare degradation behavior across materials, processes, or test conditions. In these situations, review whether the reported intervals are wide, parameters are strongly coupled, fitted parameter values are extreme or scientifically implausible, or small changes in the data or fitting settings produce substantially different estimates. Strong fit metrics do not compensate for parameters that are poorly estimated or scientifically implausible.

A.8 Considering model complexity

More complex models (i.e., those with more parameters) may fit observed data more closely, but with small datasets this may be a liability rather than an asset. A flexible model may trace the noise in the data rather than the underlying pattern, and its parameters may be poorly estimated as a result, an issue called *overfitting*. Adjusted R^2 , AIC, AIC_c, BIC, cross-validation metrics, and residual diagnostics provide complementary information about whether added complexity is justified [26].

When two models have similar fit and similar empirical support, a simpler model is often easier to justify when it is scientifically plausible and consistent with the intended use.

A.9 Scientific plausibility and intended use

Statistical performance alone is not sufficient to evaluate a model. Scientific plausibility also depends on what is known about the polymer, degradation mechanism, and test conditions. A model that is statistically competitive but

produces a fitted shape, parameter values, or extrapolated behavior that conflicts with known degradation behavior may be difficult to justify regardless of its metrics.

The intended use also affects the amount of supporting evidence and uncertainty relevant to the analysis. Exploratory model development may accommodate more uncertainty than a model intended to support a regulatory argument, a design change, or a prediction of in vivo performance.

A.10 Suggested model review workflow

1. Confirm that the dataset, preprocessing choices, and selected response values (the measured values being modeled) are appropriate for the intended analysis.
2. Review fitted curves and residual plots for each candidate model, checking for systematic patterns or implausible fitted shapes.
3. Compare goodness-of-fit metrics such as R^2 and RMSE in the context of assay variability, endpoint importance, and intended use.
4. Review cross-validation performance and fit/CV ratios to assess sensitivity to the available data.
5. Review relative model evidence using information criteria to compare fit quality against model complexity.
6. Review fitted-parameter estimates, 95% confidence intervals, uncertainty diagnostics, and scientific plausibility.
7. Consider whether model complexity is justified by the available data and the degradation mechanism.
8. Document the user-selected final model based on a balance of statistical performance, scientific plausibility, and intended use.

A.11 Questions to guide model review

The following questions are intended as practical prompts for reviewing model results, not as formal acceptance criteria.

A.11.1 Does the model describe the observed data well?

- Does the fitted curve follow the main trend in the data?
- Are performance metrics such as R^2 , RMSE, and NRMSE reasonable for the intended use?
- Is the prediction error small enough to be meaningful relative to assay variability, measurement uncertainty, or the degradation endpoint being evaluated?
- Are there regions where the model consistently overpredicts or underpredicts?

A.11.2 Does the model generalize beyond the fitted data?

- Are cross-validation metrics similar to the metrics for the model fitted to the full dataset, and are fit/CV ratios near 1?
- Could the cross-validation result be sensitive to the splitting scheme, number of observations, or a small number of influential points?
- If cross-validation performance is poor, is the model still useful as an exploratory description, or would a simpler or more stable model be more appropriate?

A.11.3 Are the model parameters interpretable and stable?

- Are the fitted parameter values scientifically plausible?
- Are the 95% confidence intervals narrow enough that the parameter estimates appear reasonably precise?

- Are any parameters strongly coupled—meaning different combinations of parameter values could produce a similar fitted curve?
- If coupling is present, does it coincide with large parameter uncertainty or an identifiability caution?
- Are individual parameter values being interpreted mechanistically, or is the model being used mainly to describe the overall curve shape?

A model can have a visually reasonable fitted curve while still having uncertain or coupled parameters. In that case, the curve may be useful for empirical description within the observed data range. Interpret individual parameter values cautiously.

A.11.4 Is the model unnecessarily complex?

- Does the more complex model improve fit, cross-validation behavior, or relative model evidence enough to justify the added parameters?
- Do AIC, AIC_c, or BIC support the additional complexity within the candidate model set?
- If two models perform similarly, is the simpler one easier to explain and justify?
- Does the dataset contain enough observations to support the number of fitted parameters?

A.11.5 Is the model scientifically plausible?

- Is the fitted curve shape consistent with expected polymer degradation behavior?
- Are the model assumptions reasonable for the material, test conditions, and degradation endpoint?
- Does the model behave plausibly near the edges of the observed data range, and would it remain plausible if used for interpolation or extrapolation?

A statistically competitive model may still be difficult to justify if its shape, parameter values, or extrapolated behavior conflict with the known degradation process.

A.11.6 Is the model appropriate for the intended use?

- Is the model being used for exploratory analysis, interpolation, prediction, comparison across conditions, or support for a regulatory rationale?
- Is the level of uncertainty acceptable for that purpose?
- Are the model's limitations clearly documented?
- Would an independent reviewer understand why this model was selected over reasonable alternatives?

Potential elements of a final model rationale include fitted performance, cross-validation behavior, relative empirical support, parameter diagnostics, residual behavior, scientific plausibility, and intended use.

A.12 Example interpretation statements

The following examples illustrate wording for possible adaptation in reports. Adapt the language to the specific analysis and supporting evidence.

- Model A had the lowest RMSE and highest adjusted R^2 among the evaluated models, indicating the strongest goodness-of-fit under the selected preprocessing conditions.
- Model A was selected over the more complex Model B because both models showed similar fit and AIC_c support (Delta AIC_c < 2), indicating that the additional complexity of Model B was not justified by the data.
- Although Model C had strong goodness-of-fit, its cross-validation error was substantially higher than its fitted error, suggesting sensitivity to which observations were included in fitting.
- Residuals for Model A were scattered around zero without obvious systematic patterns, consistent with a model form that captures the observed data.

- Model B showed a systematic trend in the residuals across time, suggesting the selected model form does not fully capture the degradation profile.
- The fitted parameters for Model A were within plausible ranges for the degradation mechanism, with narrow confidence intervals indicating stable estimation.
- Model C showed strong parameter coupling between the rate and shape parameters, with relative standard errors exceeding 100%. Parameter estimates were retained for curve description but were not interpreted individually.
- The selected model was chosen based on a balance of goodness-of-fit, cross-validation behavior, relative empirical support, parameter stability, and scientific plausibility for the degradation mechanism and intended use.

B. Worked Example: Comparing Candidate Models

B.1 Purpose of the example

This worked example illustrates how fitted curves, performance metrics, cross-validation, residual plots, Q-Q plots, and parameter uncertainty results can be considered together when comparing candidate models. It provides practical examples of residual patterns that can help readers interpret their own results; further illustrated examples are available in [24, 23, 27, 28, 29]. The dataset is simulated so the underlying relationship is known, but the comparison is presented in the same way as an analysis of an unknown experimental relationship. The example is intended as a practical demonstration, not as an acceptance procedure or a set of model-selection criteria.

An accompanying Python script generates a noisy single-exponential decay dataset and fits four candidate models. The four candidate models fitted to the primary dataset are:

- a linear model, used as an obviously inadequate form for exponential decay;
- an exponential model, which matches the form used to generate the data;
- a stretched-exponential model, which can reduce error slightly by adding a shape parameter; and
- a double-exponential model, which can reduce error further while adding parameters that may be weakly estimated and strongly coupled.

This example uses the same fitting functions, performance metrics, shuffle-split cross-validation constructor, and relative evidence calculations used by the app. The figures use a simplified layout rather than reproducing the interactive app displays exactly.

B.2 Simulated dataset and analysis settings

The simulated dataset has the form

$$y = a \exp(bt) + \varepsilon,$$

where ε is independent Gaussian noise. The random seed, parameter values, noise level, and cross-validation settings are fixed so the example is reproducible.

Setting	Value
Generating model	Exponential
True parameter a	100.000
True parameter b	-0.075
Noise standard deviation	3.000
Dataset random seed	27
CV scheme	Shuffle split
CV splits	100
CV test fraction	0.25
CV random seed	12345

B.3 Start with the fitted curves

Figure 1 shows the simulated observations with all four fitted curves. The fitted curve display is the first opportunity to identify clearly implausible behavior, but models that differ meaningfully can appear similar when the response scale is large relative to the residual differences. For this example, the linear model visibly misses the decay shape. The three exponential-family models can appear similar at the scale of the response plot, even though the stretched and double-exponential forms use additional parameters.

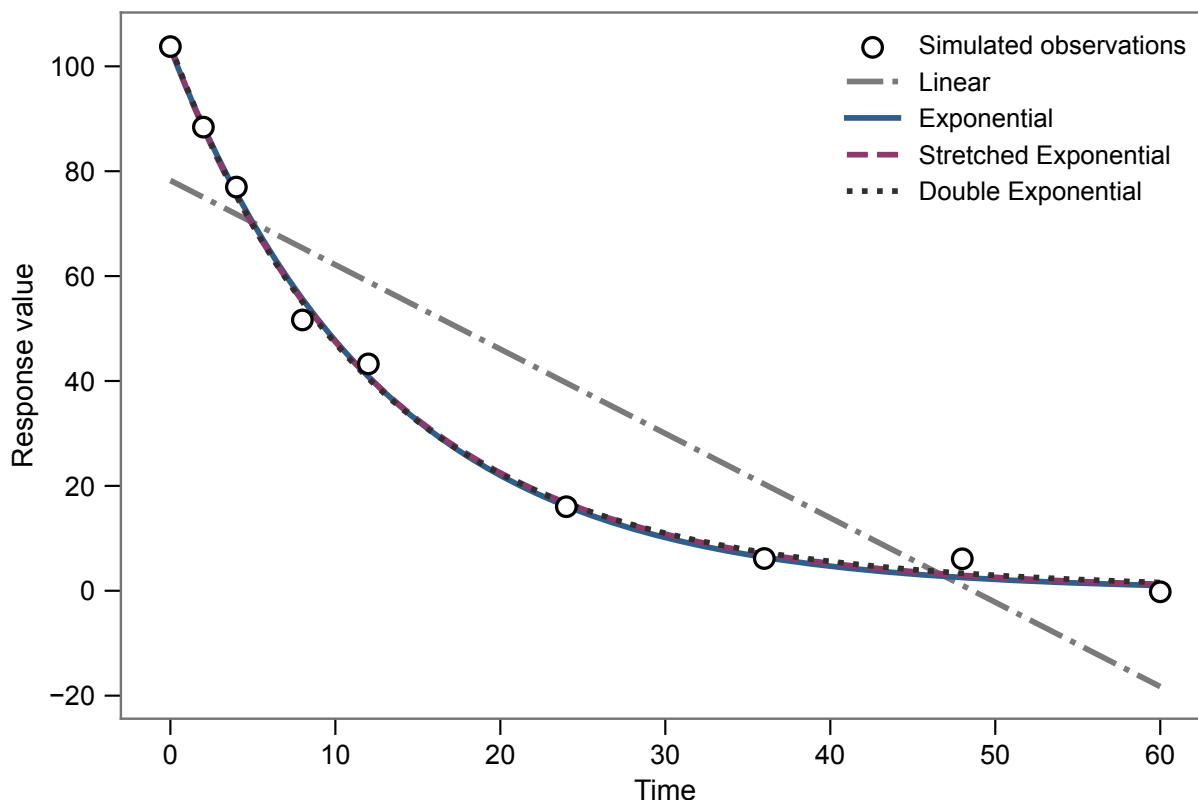


Figure 1: Candidate models fitted to the simulated dataset.

B.4 Compare fit metrics, model complexity, and cross-validation

Table 16 reports goodness-of-fit metrics, AIC_c comparisons, cross-validation error, and a compact parameter-correlation summary. These quantities answer different questions. R^2 and RMSE describe agreement with the fitted observations. AIC_c compares relative empirical support while penalizing additional fitted parameters. Cross-validation RMSE describes prediction error for omitted observations. The ratio of cross-validation RMSE to full-fit RMSE provides a simple indication of how much prediction error increases when part of the dataset is excluded from fitting.

Table 16: Model performance summary.

Model	R^2	RMSE	AIC_c	Delta AIC_c	CV RMSE	CV/fit
Linear	0.805	16.338	56.283	37.381	24.279	1.486
Exponential	0.997	2.048	18.902	0.000	2.783	1.359
Stretched Exponential	0.997	1.987	23.156	4.254	3.557	1.791
Double Exponential	0.997	1.960	30.112	11.209	1261.292	643.566

In addition to its visible lack of fit, the linear model has substantially poorer fit metrics. Its R^2 remains moderately high because it captures part of the large overall change in response, even though it does not describe the curved decay pattern well. The exponential model provides the strongest overall balance: it represents the generating relationship with only two parameters and has the lowest AIC_c . The stretched and double-exponential models reduce RMSE slightly, but their higher AIC_c values indicate that these small improvements do not offset the added complexity. Cross-validation provides a separate check on how prediction error changes when observations are omitted from fitting. The cross-validation results are similar to the full-fit results for the linear, exponential, and stretched-exponential models. In contrast, the double-exponential model gives much poorer cross-validation performance, indicating greater sensitivity to the observations used for fitting.

B.5 Use residuals versus time to look for missed structure

Residuals are calculated as observed minus predicted response (i.e., the prediction errors). A suitable fitted form generally leaves residuals distributed around zero without sustained curvature or adjacent groups that remain mainly above or below zero. Figure 2 uses the same y-axis range for all four models so the sizes and patterns can be compared directly.

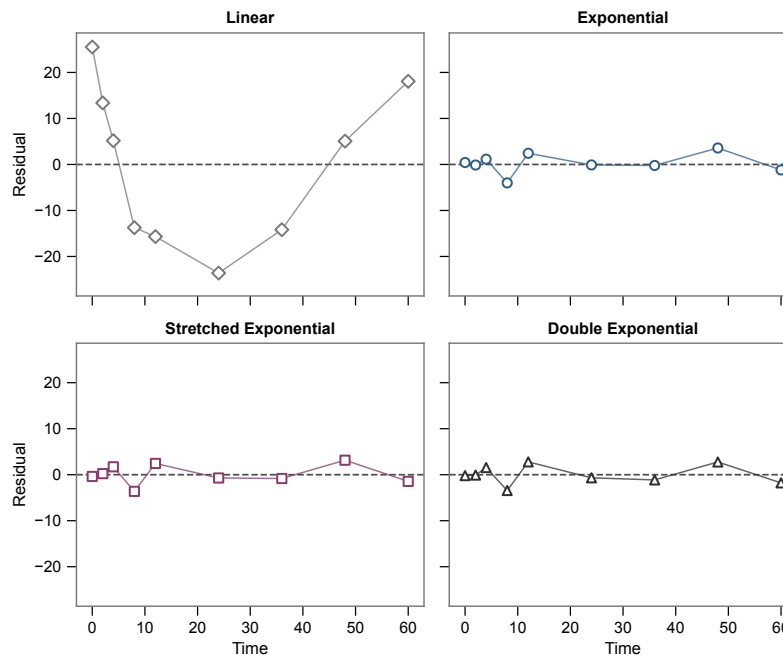


Figure 2: Residuals versus time for the four fitted models.

The linear residuals show a pronounced systematic pattern, indicating that the model does not represent the curved decay well. The exponential residuals fluctuate around zero without any sustained structure. The stretched and double-exponential models produce similarly centered residuals that are slightly smaller at some observations, but this modest improvement should be considered alongside their added complexity, AIC_c , cross-validation behavior, and parameter uncertainty.

B.6 Use Q–Q plots to review residual distribution shape

A normal Q–Q plot compares ordered standardized residuals with the quantiles expected from a normal distribution. Points close to the reference line are broadly consistent with the assumed residual distribution. Curvature, separation at the ends, or an isolated point can draw attention to skewness, heavy tails, or an unusual residual. With a small dataset, the plot has limited resolution and is interpreted together with the residual-versus-time plot rather than as a formal confirmation of normality.

These Q–Q plots do not distinguish the candidate models as clearly as the residual-versus-time plots, largely due to the small number of observations. A Q–Q plot describes the residual distribution, whereas a residual-versus-time plot is more directly informative about systematic lack of fit across the observed profile. For this dataset, the app would not display Q–Q plots for the exponential-family models because the residual variation is too small relative to the overall response range. Instead, it would report that the Q–Q diagnostic is not informative rather than magnifying very small numerical differences.

B.7 Effect of the number of observations

Figures 1–3 use the primary dataset with 9 observations. To illustrate the effect of denser sampling, a second independent dataset was generated from the same single-exponential relationship and noise distribution at 61 evenly spaced times from 0 to 60. Figure 4 shows the fitted curves and residual diagnostics for the linear and exponential

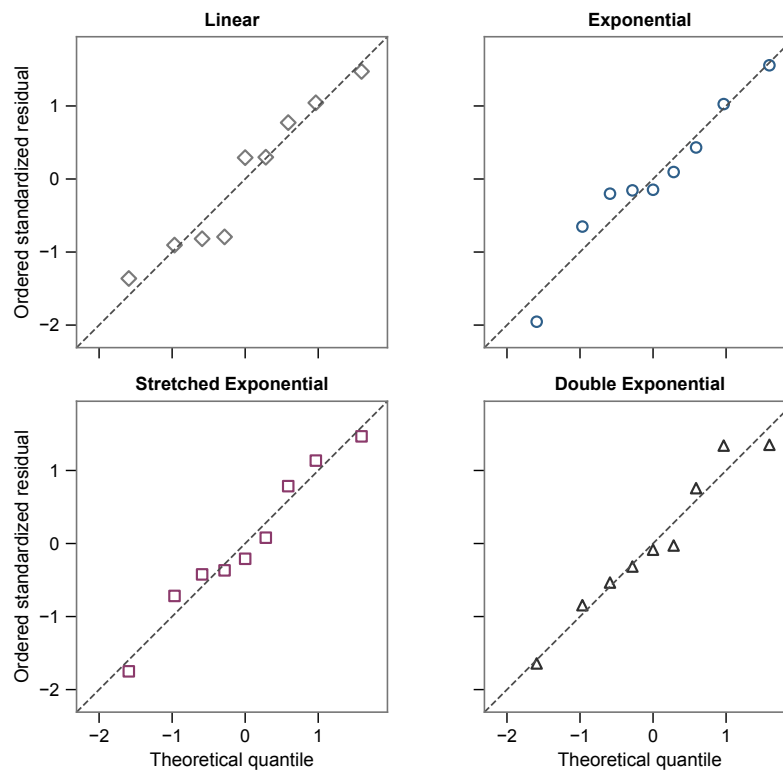


Figure 3: Normal Q-Q plots of standardized residuals.

models. This section provides a qualitative example of how the number of observations can affect the ability to distinguish a suitable model from one with systematic lack of fit.

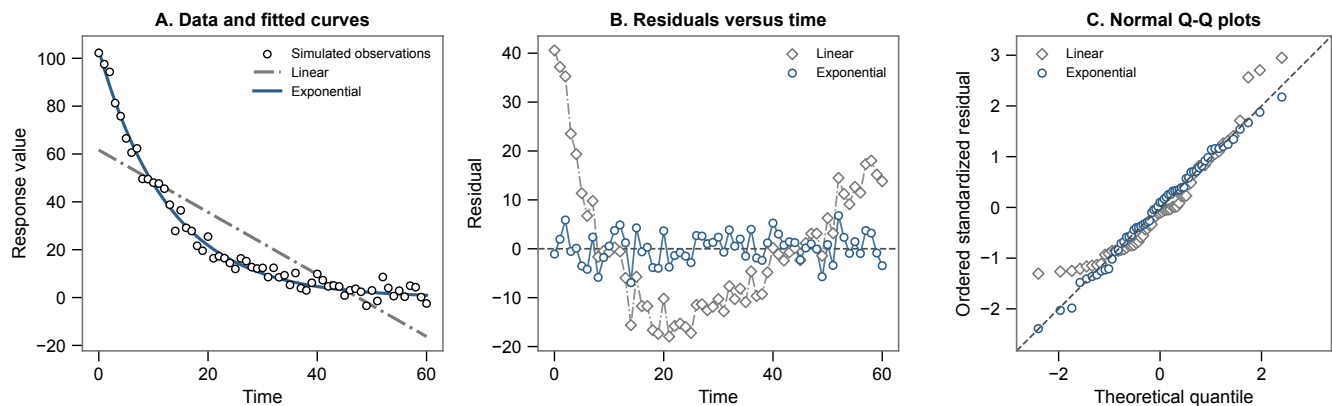


Figure 4: Graphical diagnostics for linear and exponential fits to a simulated dataset with 61 evenly spaced observations: fitted curves, residuals versus time, and normal Q-Q plots.

The smaller dataset already suggests that the linear model does not capture the decay relationship, but the denser dataset makes the lack of fit easier to diagnose. With more observations, the linear residuals form a clearer systematic pattern over time rather than fluctuating randomly around zero, and departures from the Q-Q reference line are easier to assess. In contrast, the exponential-model residuals remain comparatively patternless around zero and follow the Q-Q reference line more closely. Additional observations do not guarantee selection of a suitable model, but they provide more information for distinguishing systematic lack of fit from random variation. With sparse data, the same patterns may be less apparent or more easily mistaken for noise.

B.8 Review fitted-parameter uncertainty

The app reports each fitted parameter with a 95% confidence interval. Table 17 also reports the standard error (SE) and relative standard error (RSE = SE / estimate in %).

Table 17: Fitted-parameter estimates and approximate uncertainty.

Model	Parameter	Estimate	SE	95% CI	RSE (%)
Linear	a	78.242	8.936	21.131	11.421
Linear	b	-1.609	0.300	0.709	18.628
Exponential	a	103.344	1.758	4.157	1.701
Exponential	b	-0.077	0.003	0.008	4.257
Stretched Exponential	a	104.141	2.281	5.580	2.190
Stretched Exponential	b	-0.085	0.014	0.034	16.204
Stretched Exponential	c	0.964	0.057	0.139	5.874
Double Exponential	a	64.400	355.320	913.379	551.739
Double Exponential	b	-0.096	0.156	0.401	161.667
Double Exponential	c	39.562	356.112	915.415	900.144
Double Exponential	d	-0.056	0.133	0.343	238.297

Table 18 reports the strongest pairwise parameter correlation for each fitted model. Strong correlation can indicate that different combinations of parameter values produce similar fitted curves, especially in flexible nonlinear models. The app considers correlation together with relative standard error: high correlation alone is not unusual, but high correlation combined with large parameter uncertainty provides stronger evidence that the model parameters cannot be uniquely identified.

Table 18: Largest absolute pairwise parameter correlation for each fitted model.

Model	Maximum absolute correlation	Parameter pair
Linear	0.723	a-b
Exponential	0.604	a-b
Stretched Exponential	0.963	b-c
Double Exponential	1.000	a-c

The exponential parameters are expected to be estimated comparatively precisely because the fitted model matches the generating relationship, and indeed their relative standard errors are only a few percent. For the stretched exponential, the estimated shape parameter remains close to the single-exponential value of 1, so the added parameter changes the fitted curve only slightly and does not improve AIC_c . In the double-exponential fit, both components retain nonzero amplitudes (a and c), but the amplitudes and rate parameters (b and d) have very large uncertainty and are almost perfectly coupled. The fitting procedure can trade one component against the other while producing nearly the same overall curve. In the app, this combination of high parameter correlation and elevated relative standard errors would receive an *identifiability caution* badge. This illustrates why a slightly closer fitted curve does not necessarily support the added model structure or interpretation of the individual component parameters.

B.9 Integrated interpretation

No single output determines the preferred candidate. In this example, the linear model is readily set aside because the fitted curve and residual pattern show clear lack of fit. The exponential model provides the strongest overall balance: it captures the curve shape without obvious residual structure and does so with the fewest parameters among the plausible candidates. The stretched and double-exponential models can achieve slightly lower error, but their higher AIC_c values indicate that the improvement does not justify the added complexity for this dataset. The double-exponential parameter results provide an additional caution because both fitted components remain active while their amplitudes and rates are extremely uncertain and strongly coupled.

For an experimental analysis, the same workflow can be applied without knowing the generating model: inspect the fitted curves, compare complementary metrics, review cross-validation as a sensitivity diagnostic, examine residual-versus-time and Q-Q plots, and consider parameter uncertainty and scientific plausibility before documenting the user-selected model.

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